

Academic Openness Index: Methodology and Formulas

The dataset consists of cross-country observations capturing key dimensions of academic system openness. The following indicators are used:

- Mobile doctorate graduates (Mobile_dct_grad)
- International net flows of scientific authors (Net_flows)
- Percentage of international scientific collaboration (pct_int_coll)
- Total number of researchers (Total_researchers)

All variables refer to the most recent available year for each country. To ensure cross-country comparability, absolute counts are normalized by the total number of researchers. This controls for differences in country size and research system scale.

Formulas:

$$Mobile_per_1000_i = \left(\frac{Mobile_dct_grad_i}{Total_researchers_i} \right) / 1000$$
$$Flows_per_1000_i = \left(\frac{Net_flows_i}{Total_researchers_i} \right) / 1000$$

The international collaboration variable is already expressed as a percentage and is therefore not normalized. To combine variables measured in different units, all indicators are standardized using z-scores:

$$Z_{k,i} = \frac{(X_{k,i} - \bar{X}_k)}{sd(X_k)}$$

where k indexes indicators and i indexes countries. The Academic Openness Index is constructed as the average of standardized indicators:

$$Openness_i = \frac{Z_mobile_i + Z_flows_i + Z_collab_i}{3}$$

This approach assigns equal weight to each dimension of academic openness. Higher values of the index indicate more open academic systems characterized by:

- greater researcher mobility
- positive international flows
- higher levels of international collaboration

Lower values indicate more closed systems, often associated with academic inbreeding and limited external circulation. The index can be related to macroeconomic outcomes such as GDP per capita:

$$Openness_i = \alpha + \beta \times \log GDPpc_i + \varepsilon_i$$

This allows testing whether academic openness is associated with economic development.